AWS COST MONITORING & BUDGET ALERTS

Services: AWS Budgets, Cost Explorer, Lambda, SES

Goal: Automatically notify when budget thresholds are met.

Skills: Billing alerts, email notifications, tagging strategies.

1. Definition

AWS Cost Monitoring & Budget Alerts is a cloud cost management project that automatically monitors AWS spending and sends alerts when the spending reaches defined budget thresholds. The system uses AWS services to detect when the budget limit is approaching or exceeded and sends email notifications to inform the user. This helps prevent unexpected cloud bills and improves cost control.

2. Scope of the Project

The main scope of this project is to monitor AWS usage costs and notify users automatically when the spending crosses a defined limit.

The project includes:

- Creating a monthly AWS budget
- Setting alert thresholds (85% and 100%)
- Sending notifications using AWS SNS
- Triggering an AWS Lambda function when alerts occur
- Sending automated email alerts using Amazon SES

This system helps individuals or organizations track their AWS spending and take action before exceeding their budget.

3. Methodologies

This project follows an **event-driven architecture**.

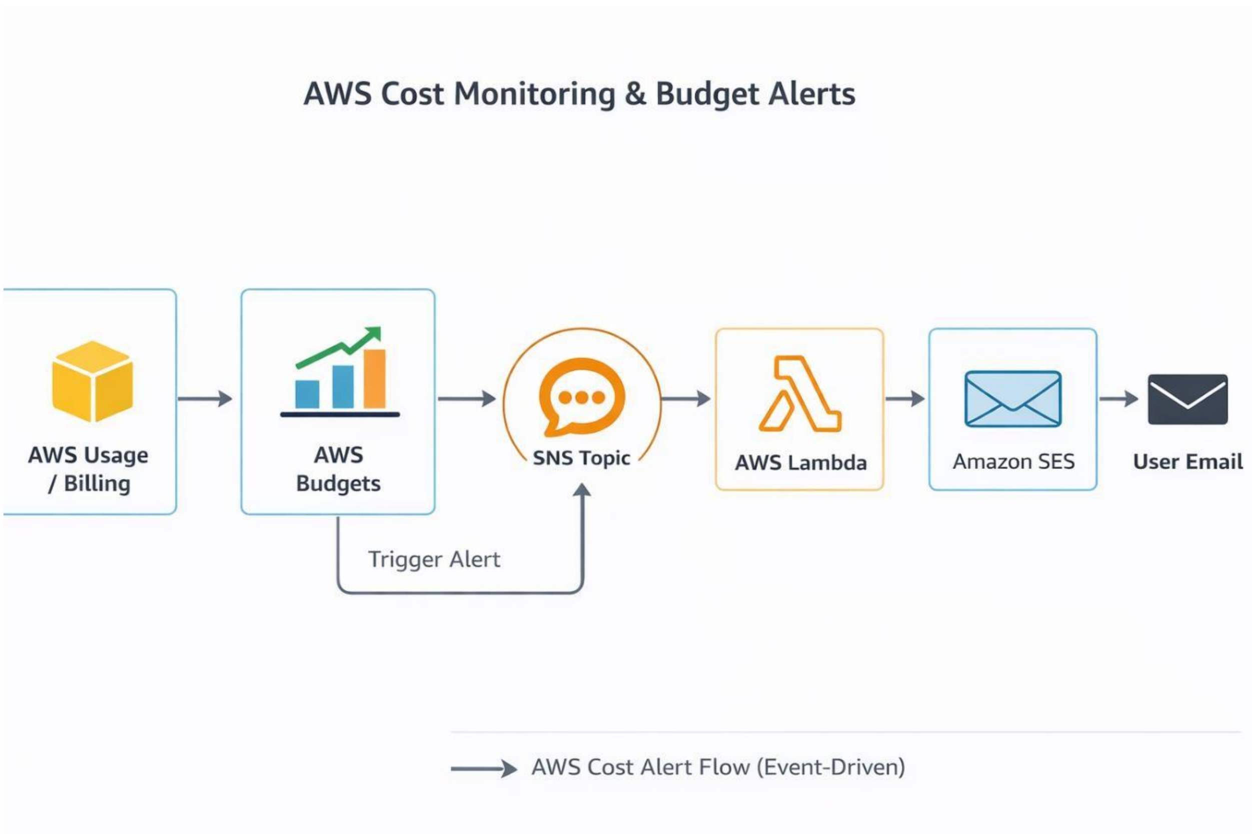
Steps in the methodology:

1. Monitor AWS spending using **AWS Budgets**
2. When spending crosses a defined threshold, an alert is generated

3. The alert is sent to **Amazon SNS**
4. SNS triggers an **AWS Lambda function**
5. Lambda processes the event and sends an email using **Amazon SES**

This approach ensures automated monitoring and notification without manual intervention.

4. Architecture Diagram / Flow



5. Explanation of Services Used

AWS Budgets

AWS Budgets allows users to set spending limits for AWS services. It monitors the cost usage and triggers alerts when the spending reaches the defined threshold.

Purpose:

- Monitor AWS spending
- Generate alerts when budget limits are reached

Amazon SNS (Simple Notification Service)

Amazon SNS is a messaging service used to send notifications and trigger other services.

Purpose:

- Receive alerts from AWS Budgets
- Forward alerts to AWS Lambda

AWS Lambda

AWS Lambda is a serverless compute service that runs code automatically in response to events.

Purpose:

- Process the alert notification
- Trigger email sending through SES

Amazon SES (Simple Email Service)

Amazon SES is an email service used to send emails from AWS applications.

Purpose:

- Send automated email alerts to users when budget thresholds are reached

6. Step-by-Step Procedure

Step 1: Create AWS Budget

1. Login to AWS Console.
2. Navigate to **Billing → Budgets**.
3. Click **Create Budget**.
4. Select **Monthly Cost Budget**.
5. Enter the budget amount (example: $10).
6. Configure alerts for:

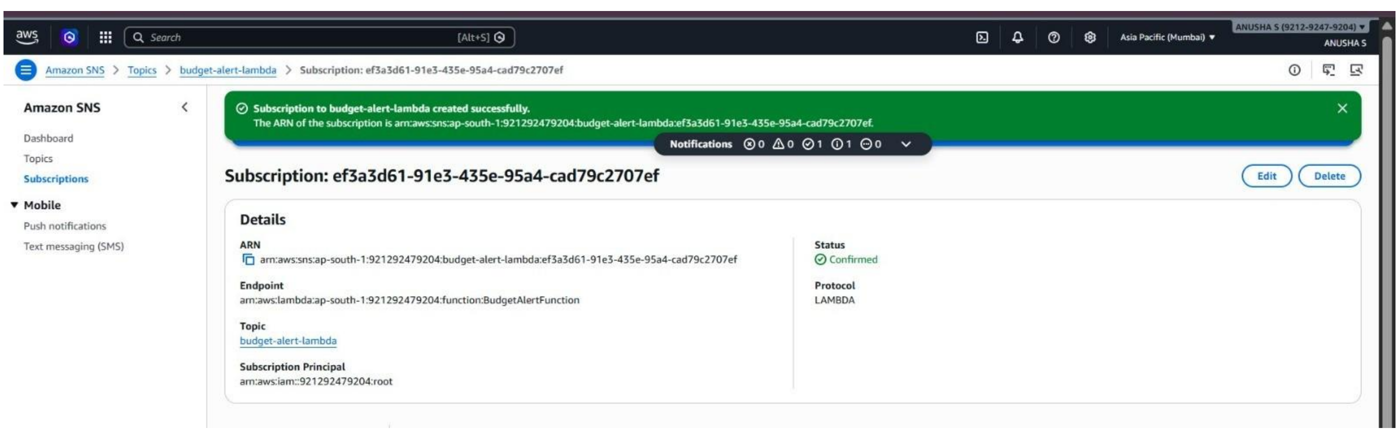
- 85% threshold
- 100% threshold.

Step 2: Configure Email Notification

1. Enter the email address to receive alerts.
2. Configure the threshold percentage for alerts.
3. Save the budget configuration.

Step 3: Create SNS Topic

1. Go to **Amazon SNS** service.
2. Click **Topics → Create Topic**.
3. Select **Standard Topic**.
4. Name the topic (example: budget-alert-topic).
5. Create the topic.

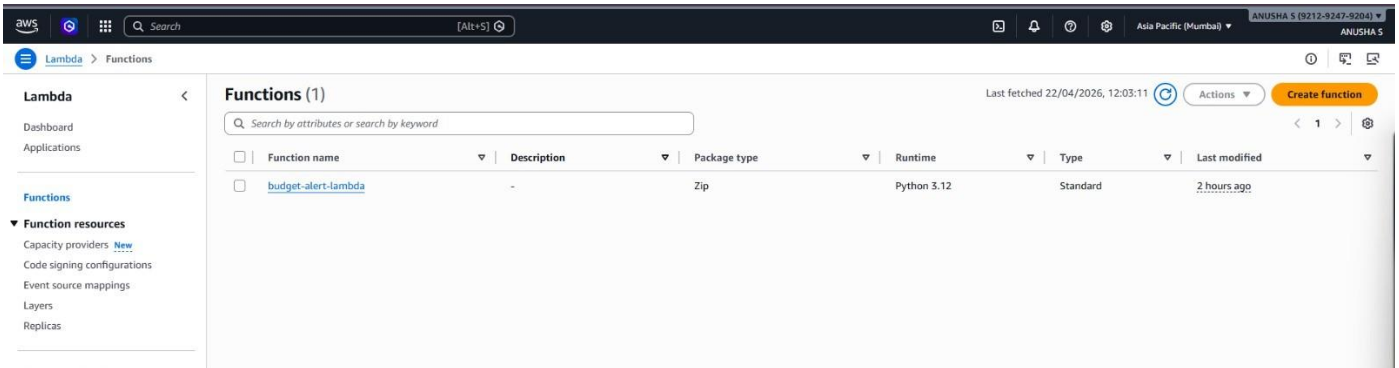


Step 4: Create Lambda Function

1. Navigate to **AWS Lambda**.
2. Click **Create Function**.
3. Choose **Author from Scratch**.

4. Select runtime **Python 3.x**.

5. Create the Lambda function.

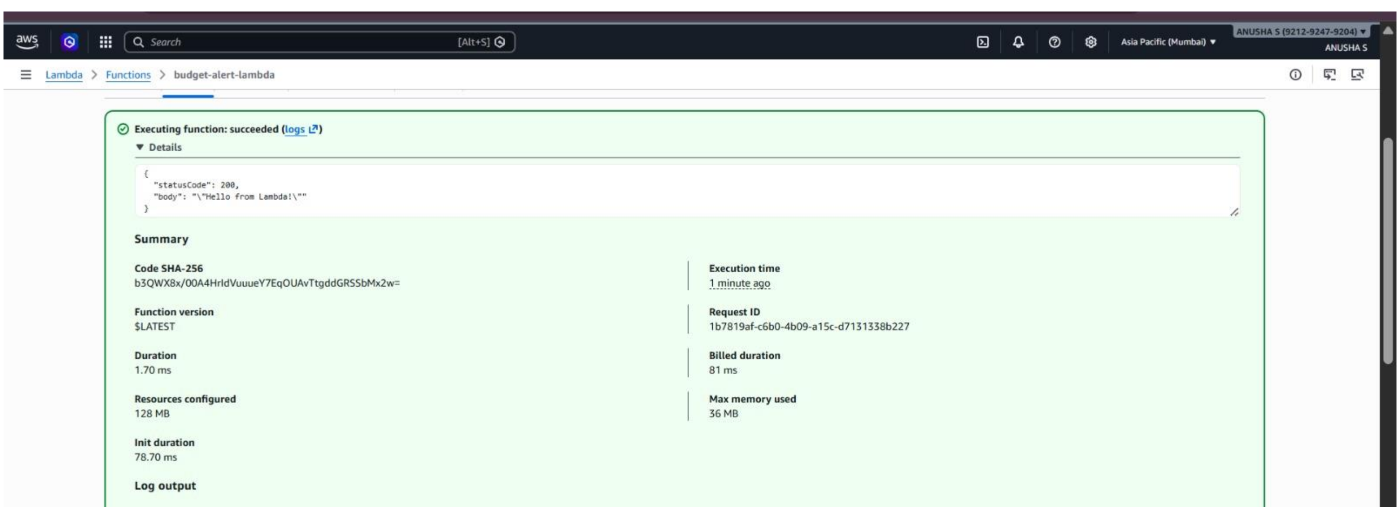


Step 5: Add Lambda Code

The Lambda function is used to send an email alert through Amazon SES when triggered.

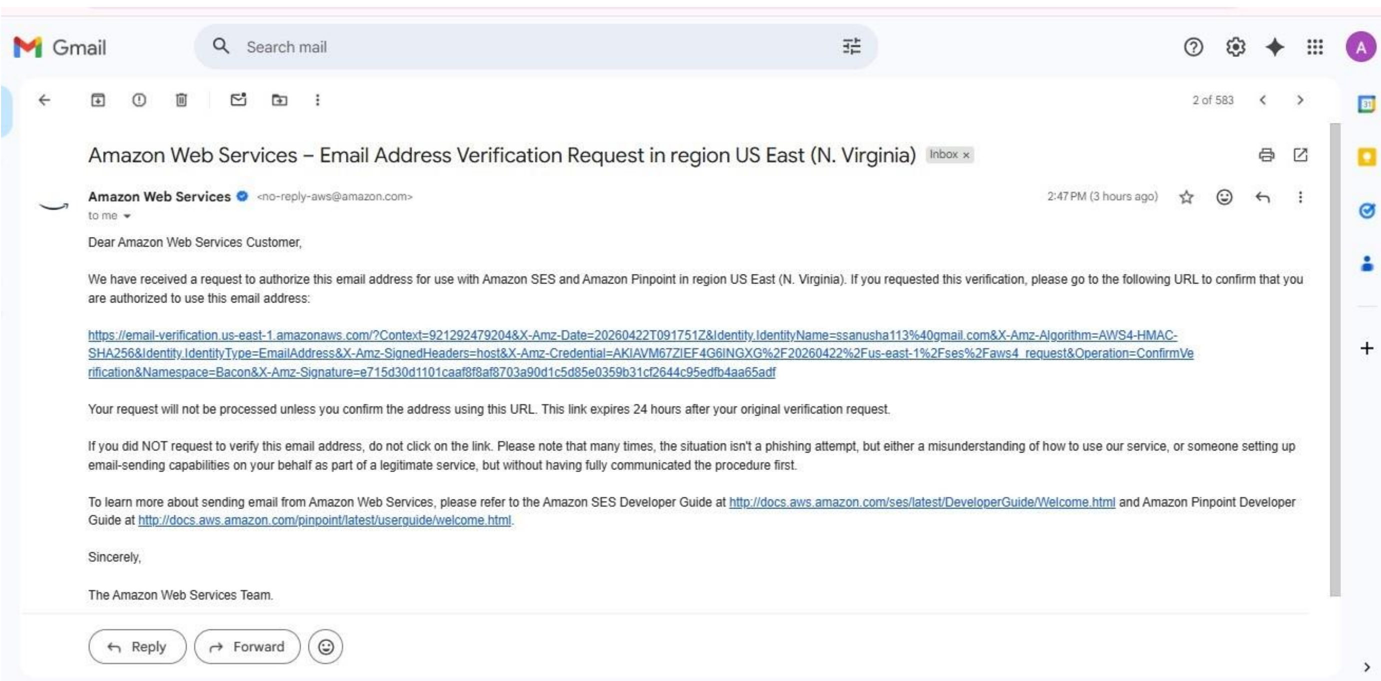
Example logic:

- Receive alert from SNS
- Process event
- Send email notification using SES



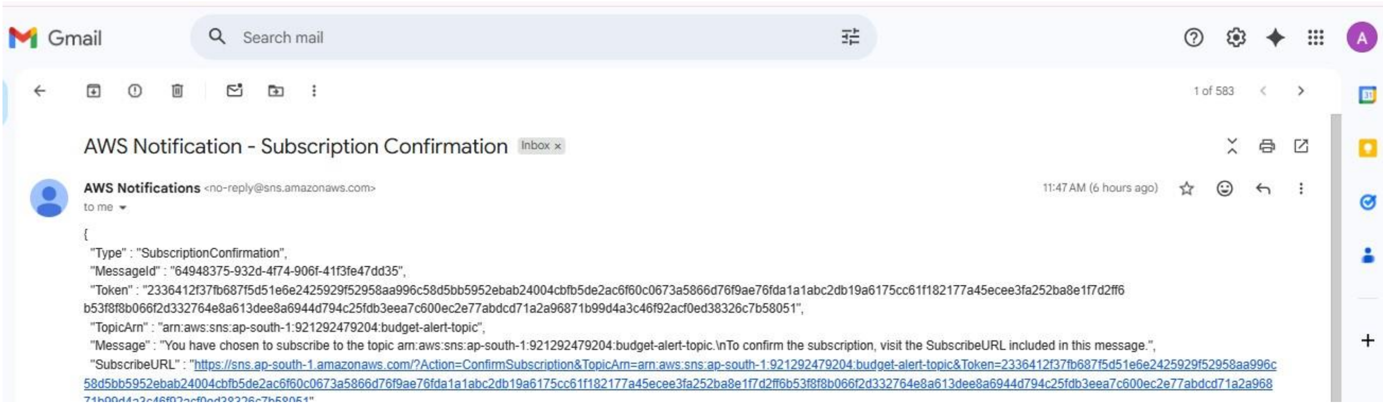
Step 6: Verify Email in SES

1. Go to **Amazon SES**.
2. Click **Identities**.
3. Add your email address.
4. Verify the email by clicking the confirmation link sent to your inbox.



Step 7: Subscribe Lambda to SNS

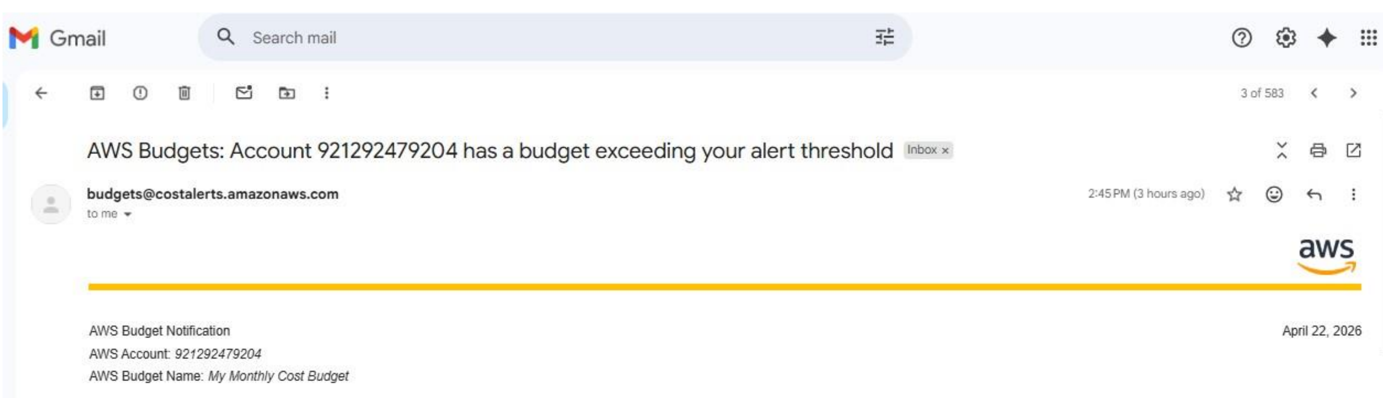
1. Open SNS topic.
2. Click **Create Subscription**.
3. Select protocol **AWS Lambda**.
4. Choose the created Lambda function.

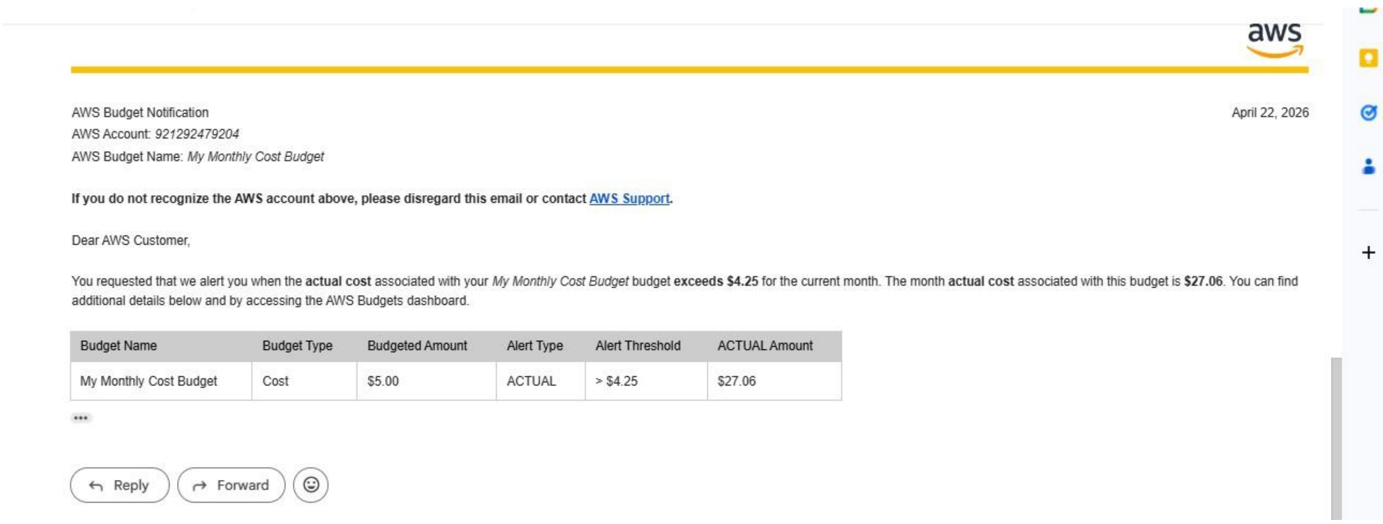


Step 8: Connect Budget to SNS

1. Go to **AWS Budgets**.
2. Edit the budget alerts.
3. Expand **Amazon SNS Alerts**.
4. Paste the SNS topic ARN.
5. Save the changes.

Now the automated alert system is active.





aws

AWS Budget Notification
AWS Account: *921292479204*
AWS Budget Name: *My Monthly Cost Budget*

April 22, 2026

If you do not recognize the AWS account above, please disregard this email or contact AWS Support.

Dear AWS Customer,

You requested that we alert you when the **actual cost** associated with your *My Monthly Cost Budget* budget **exceeds $4.25** for the current month. The month **actual cost** associated with this budget is **$27.06**. You can find additional details below and by accessing the AWS Budgets dashboard.

Budget Name	Budget Type	Budgeted Amount	Alert Type	Alert Threshold	ACTUAL Amount
My Monthly Cost Budget	Cost	$5.00	ACTUAL	> $4.25	$27.06

Reply Forward

7. Troubleshooting

Issue	Method	Solution
Email not received	Check SES identity verification	Verify the email in SES
Lambda execution failed	Check CloudWatch logs	Fix permission or code issues
SNS not triggering Lambda	Check subscription status	Ensure subscription is confirmed
Budget alert not triggered	Check threshold configuration	Adjust budget amount and thresholds
Email received in spam	Check Gmail spam folder	Mark email as "Not Spam"

8. Conclusion

The AWS Cost Monitoring & Budget Alerts project provides an automated solution for monitoring AWS expenses. By integrating AWS Budgets, SNS, Lambda, and SES, the system automatically detects when spending reaches a defined threshold and sends email alerts to users. This helps prevent unexpected cloud costs and ensures better financial control over AWS resources. The project demonstrates the use of serverless architecture and event-driven automation for efficient cloud cost management.